IMPORTING LIBRARIES

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

LOADING DATA

In [2]:
```
df=pd.read_csv('sales_data_sample.csv',encoding="cp1252")
df.head()
```

Out[2]:

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	ORDER
0	10107	30	95.70	2	2871.00	2/24
1	10121	34	81.35	5	2765.90	5/7
2	10134	41	94.74	2	3884.34	7/1
3	10145	45	83.26	6	3746.70	8/25
4	10159	49	100.00	14	5205.27	10/10

5 rows × 25 columns

FINDING ITEMS STATUS

In [3]:
```
df['STATUS'].value_counts()
```

Out[3]:
```
STATUS
Shipped       2617
Cancelled       60
Resolved        47
On Hold         44
In Process      41
Disputed        14
Name: count, dtype: int64
```

KEEPING ONLY COMPLETED ITEMS AND REMOVING ONHOLD,PENDING AND DISPUTED ITEMS

In [4]:
```
df = df[df['STATUS'].isin(['Shipped', 'Resolved'])]
df.reset_index(drop=True, inplace=True)
df['STATUS'].value_counts()
```

Out[4]:
```
STATUS
Shipped     2617
Resolved      47
Name: count, dtype: int64
```

In [5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2664 entries, 0 to 2663
Data columns (total 25 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ORDERNUMBER       2664 non-null   int64
 1   QUANTITYORDERED   2664 non-null   int64
 2   PRICEEACH         2664 non-null   float64
 3   ORDERLINENUMBER   2664 non-null   int64
 4   SALES             2664 non-null   float64
 5   ORDERDATE         2664 non-null   object
 6   STATUS            2664 non-null   object
 7   QTR_ID            2664 non-null   int64
 8   MONTH_ID          2664 non-null   int64
 9   YEAR_ID           2664 non-null   int64
 10  PRODUCTLINE       2664 non-null   object
 11  MSRP              2664 non-null   int64
 12  PRODUCTCODE       2664 non-null   object
 13  CUSTOMERNAME      2664 non-null   object
 14  PHONE             2664 non-null   object
 15  ADDRESSLINE1      2664 non-null   object
 16  ADDRESSLINE2      289 non-null    object
 17  CITY              2664 non-null   object
 18  STATE             1263 non-null   object
 19  POSTALCODE        2588 non-null   object
 20  COUNTRY           2664 non-null   object
 21  TERRITORY         1646 non-null   object
 22  CONTACTLASTNAME   2664 non-null   object
 23  CONTACTFIRSTNAME  2664 non-null   object
 24  DEALSIZE          2664 non-null   object
dtypes: float64(2), int64(7), object(16)
memory usage: 520.4+ KB
```

In [6]:
```
df.duplicated().sum()
```

Out[6]:
```
np.int64(0)
```

cleanning data set and removing unwanted datas

In [7]:
```
df=df.drop(columns=['ORDERNUMBER','ORDERLINENUMBER','PRODUCTCODE','CUSTOMERNAME','P
df.head()
```

Out[7]:

	QUANTITYORDERED	PRICEEACH	SALES	ORDERDATE	STATUS	QTR_ID	MONTH_ID	Y
0	30	95.70	2871.00	2/24/2003 0:00	Shipped	1	2	
1	34	81.35	2765.90	5/7/2003 0:00	Shipped	2	5	
2	41	94.74	3884.34	7/1/2003 0:00	Shipped	3	7	
3	45	83.26	3746.70	8/25/2003 0:00	Shipped	3	8	
4	49	100.00	5205.27	10/10/2003 0:00	Shipped	4	10	

In [8]:
```
df.to_csv("clean_sales_data.csv", index=False)
```

In [9]:
```
df.describe().drop(['QTR_ID','MONTH_ID','YEAR_ID'],axis=1)
```

Out[9]:

	QUANTITYORDERED	PRICEEACH	SALES	MSRP
count	2664.000000	2664.000000	2664.000000	2664.000000
mean	34.933559	83.644208	3544.376637	100.894895
std	9.362665	20.238198	1819.719663	40.439059
min	6.000000	27.220000	553.950000	33.000000
25%	27.000000	68.647500	2204.012500	68.000000
50%	34.000000	95.905000	3182.250000	99.000000
75%	43.000000	100.000000	4492.477500	127.000000
max	97.000000	100.000000	12536.500000	214.000000

In [10]:
```
df['ORDERDATE'] = pd.to_datetime(df['ORDERDATE'])
df['YEAR_ID'] = df['ORDERDATE'].dt.year
df['MONTH_ID'] = df['ORDERDATE'].dt.month_name()
df['QTR_ID'] = df['ORDERDATE'].dt.quarter
```

KPI

TOTAL SALES

In [11]:
```
print("TOTAL SALES = ",df.SALES.sum(), "\n")
total_sales = df.groupby('PRODUCTLINE')['SALES'].sum()
for product, sales in total_sales.items():
    print(f"{product} = {sales}")
```

```
TOTAL SALES =  9442219.36

Classic Cars = 3727559.67
Motorcycles = 1129573.83
Planes = 900999.49
Ships = 631036.47
Trains = 215352.57
Trucks and Buses = 1064570.14
Vintage Cars = 1773127.19
```

TOTAL ORDER

In [12]:
```
print("TOTAL QUANTITYORDERED = ", df.QUANTITYORDERED.sum(), "\n")
total_orders = df.groupby('PRODUCTLINE')['QUANTITYORDERED'].sum()
for product, orders in total_orders.items():
    print(f"{product} = {orders}")
```

```
TOTAL QUANTITYORDERED =  93063

Classic Cars = 32249
Motorcycles = 11355
Planes = 9831
Ships = 7196
Trains = 2622
Trucks and Buses = 10212
Vintage Cars = 19598
```

Average order values

In [13]:
```
avg_order_value = total_sales / total_orders
avg_order_value
```

Out[13]:
```
PRODUCTLINE
Classic Cars        115.586830
Motorcycles          99.478100
Planes               91.648814
Ships                87.692672
Trains               82.132941
Trucks and Buses    104.246978
Vintage Cars         90.474905
dtype: float64
```

DEALSIZE DATA ANALYSIS

In [14]:
```
df['DEALSIZE'].value_counts()
```

Out[14]:
```
DEALSIZE
Medium    1304
Small     1216
Large      144
Name: count, dtype: int64
```

AVERAGE DEAL SIZES

In [15]:
```
df.groupby('DEALSIZE')['SALES'].sum()
```

Out[15]:
```
DEALSIZE
Large     1189279.10
Medium    5737563.42
Small     2515376.84
Name: SALES, dtype: float64
```

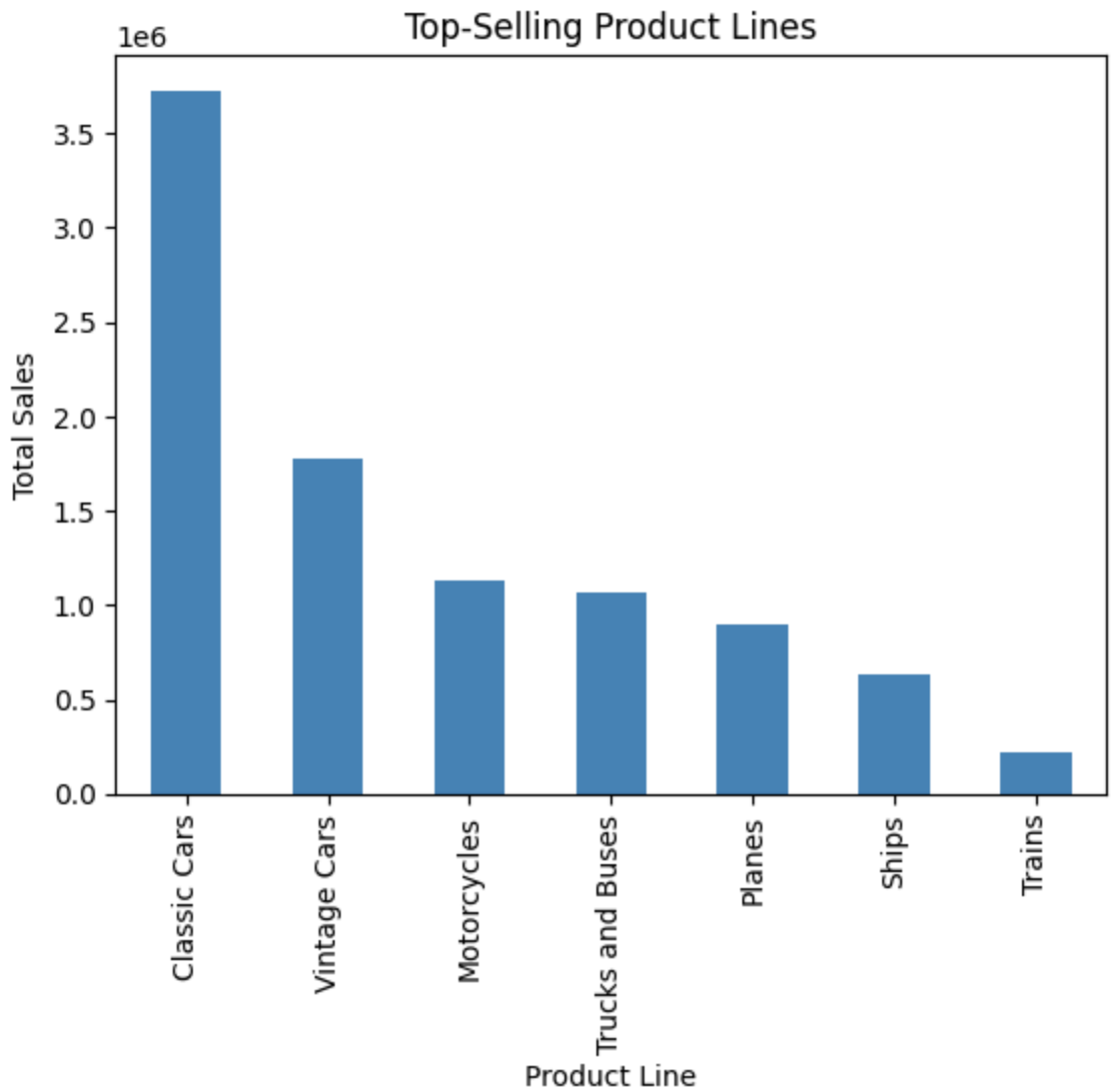
FINDING TOP PRODUCTS IN SALES

In [16]:
```
top_products = df.groupby('PRODUCTLINE')['SALES'].sum().sort_values(ascending=False
top_products
```

Out[16]:
```
PRODUCTLINE
Classic Cars        3727559.67
Vintage Cars        1773127.19
Motorcycles         1129573.83
Trucks and Buses    1064570.14
Planes               900999.49
Ships                631036.47
Trains               215352.57
Name: SALES, dtype: float64
```

BARGRAPH FOR TOP SELLING PRODUCT

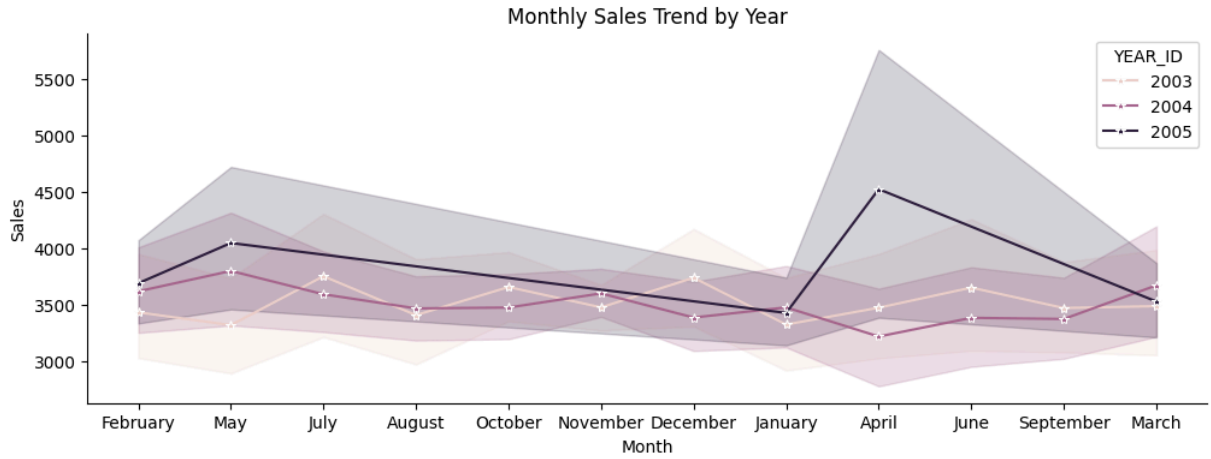
In [17]:
```
top_products.plot(kind='bar', color='steelblue')
plt.title("Top-Selling Product Lines")
plt.xlabel("Product Line")
plt.ylabel("Total Sales")
plt.show()
```



PLOTTING GRAPH FOR MONTHLY SALES WITH YEAR

In [18]:
```
plt.figure(figsize=(12, 4))
sns.lineplot(data=df, x='MONTH_ID', y='SALES', hue='YEAR_ID', marker='*')
plt.title("Monthly Sales Trend by Year")
plt.xlabel("Month")
plt.ylabel("Sales")

sns.despine()
plt.show()
```



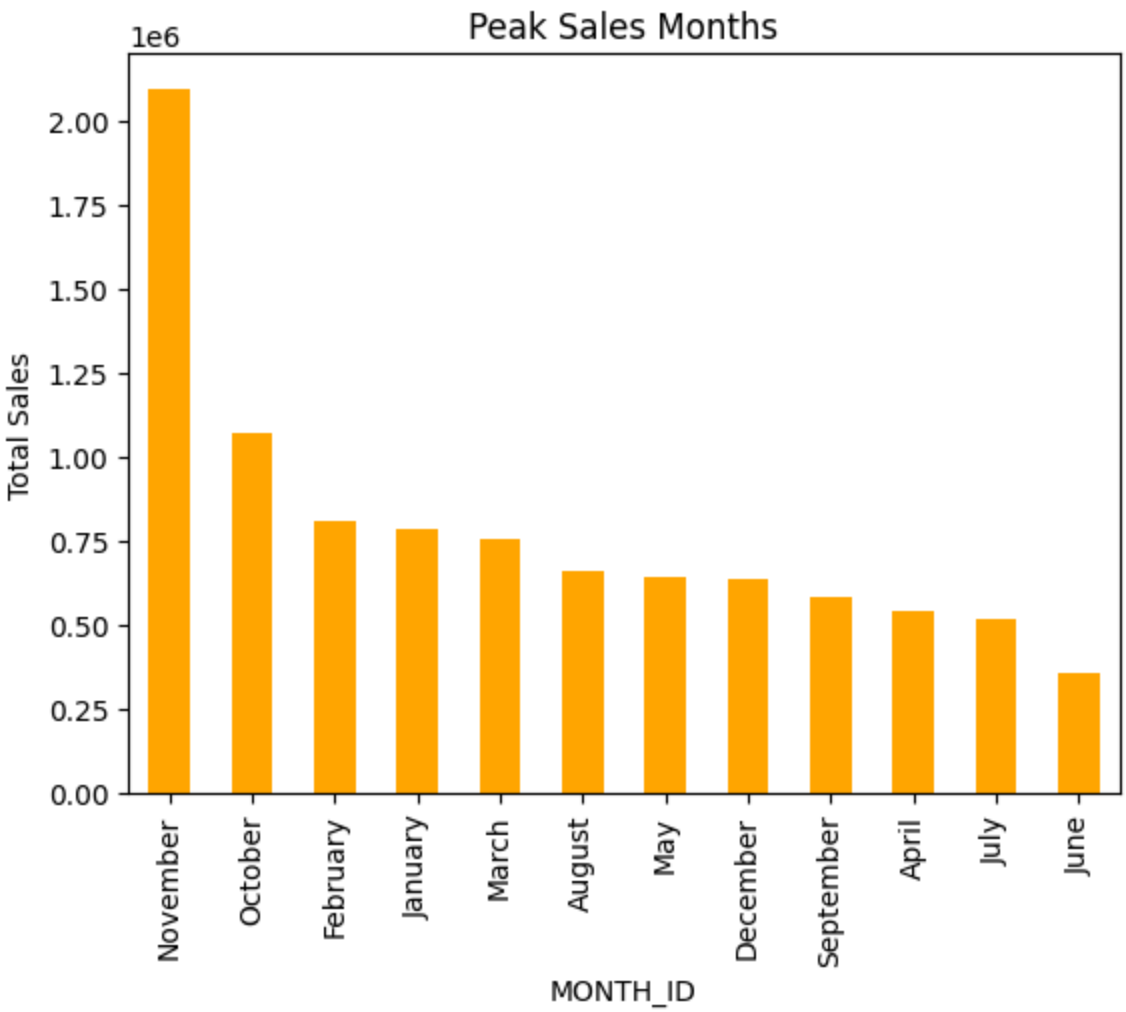
FINDING PEAK MONTHS FOR SALES

In [19]:
```
peak_months = df.groupby('MONTH_ID')['SALES'].sum().sort_values(ascending=False)
peak_months
```

Out[19]:
```
MONTH_ID
November     2092625.46
October      1072504.30
February      810441.90
January       785874.44
March         754501.39
August        659310.57
May           639368.06
December      634679.12
September     584724.27
April         538976.00
July          514875.97
June          354337.88
Name: SALES, dtype: float64
```

BARGRAPH FOR PEAK MONTH OF SALE

In [20]:
```
peak_months.plot(kind='bar', color='orange')
plt.title("Peak Sales Months")
plt.ylabel("Total Sales")
plt.show()
```



FINDING PEAK QUARTER ACCORDING TO SALES

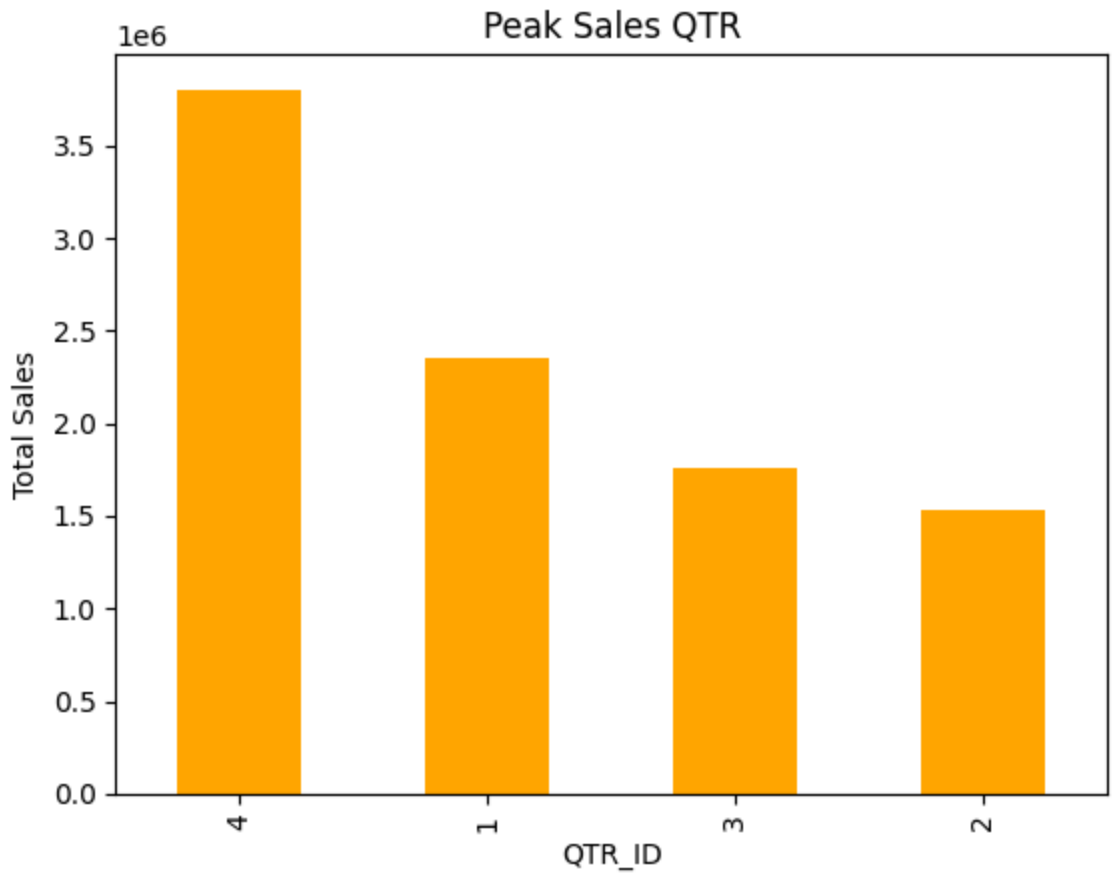
In [21]:
```
peak_QTR = df.groupby('QTR_ID')['SALES'].sum().sort_values(ascending=False)
peak_QTR
```

Out[21]:
```
QTR_ID
4    3799808.88
1    2350817.73
3    1758910.81
2    1532681.94
Name: SALES, dtype: float64
```

BAR GRAPH FOR PEAK SALES AS OF QUARTER

In [22]:
```
peak_QTR.plot(kind='bar', color='orange')
plt.title("Peak Sales QTR")
```

```
plt.ylabel("Total Sales")
plt.show()
```



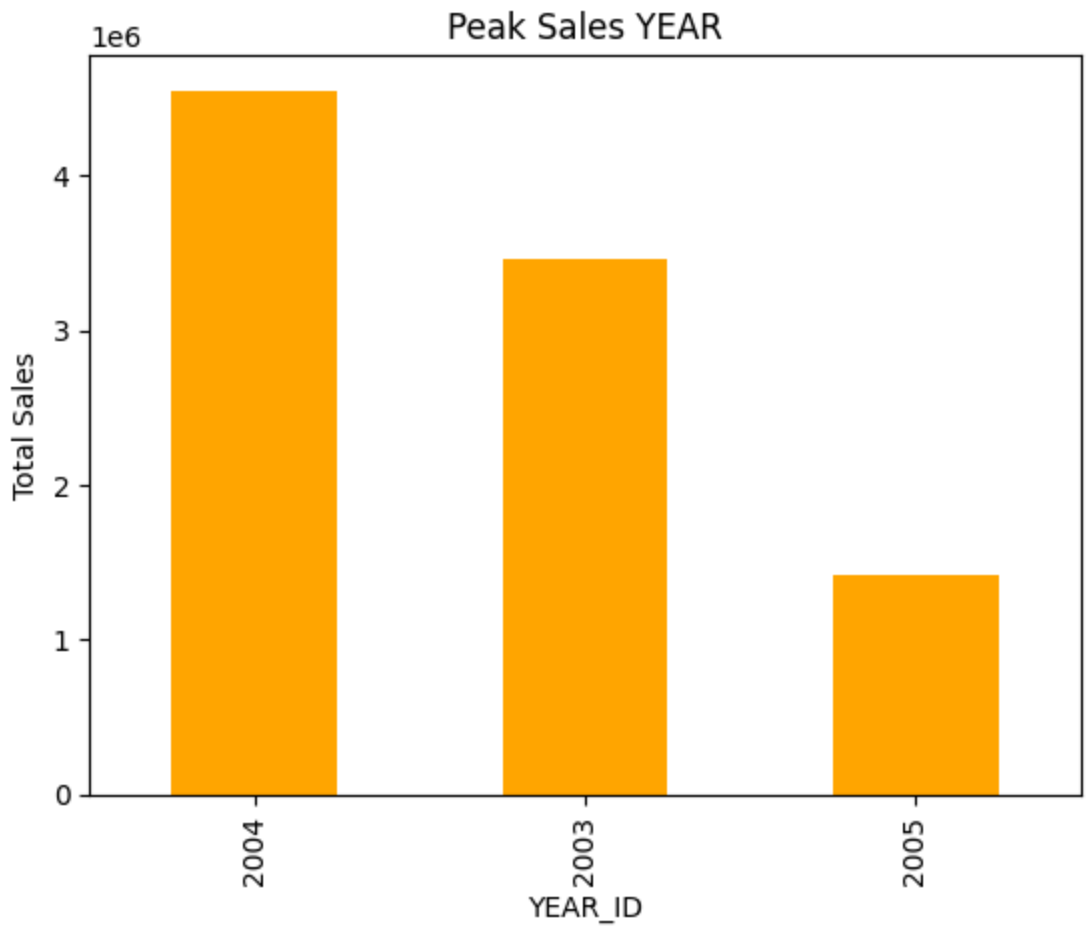
PEAK SALES AS OF YEAR

In [23]:
```
peak_YEAR = df.groupby('YEAR_ID')['SALES'].sum().sort_values(ascending=False)
peak_YEAR
```

Out[23]:
```
YEAR_ID
2004    4552125.83
2003    3468268.62
2005    1421824.91
Name: SALES, dtype: float64
```

PLOTTING SALES AS OF PEAK YEAR

In [24]:
```
peak_YEAR.plot(kind='bar', color='orange')
plt.title("Peak Sales YEAR")
plt.ylabel("Total Sales")
plt.show()
```



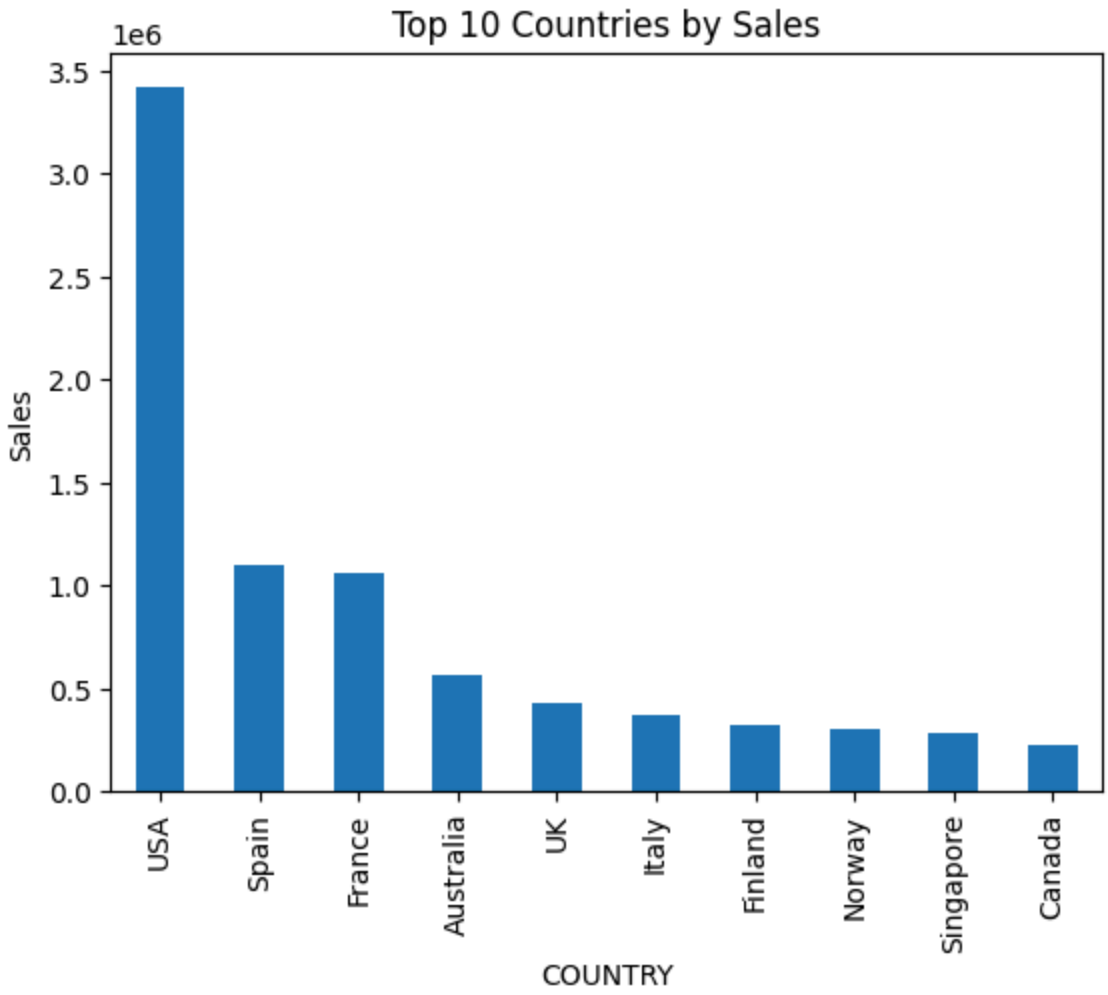
SALES AS OF PEAK COUNTRIES

In [25]:
```
country_sales = df.groupby('COUNTRY')['SALES'].sum().sort_values(ascending=False).h
country_sales
```

Out[25]:
```
COUNTRY
USA          3416477.64
Spain        1098721.03
France       1067131.83
Australia     572273.58
UK            428472.21
Italy         374674.31
Finland       329581.91
Norway        307463.70
Singapore     288488.41
Canada        224078.56
Name: SALES, dtype: float64
```

PLOTTING ACCORDING AS OF PEAK COUNTRIES

In [26]:
```
country_sales.plot(kind='bar')
plt.title("Top 10 Countries by Sales")
plt.ylabel("Sales")
plt.show()
```



FINDING CORRELATION BETWEEN SALE,QUANTITY_ORDER, PRICE_OF_EACH_PRODUCT

In [27]:
```
corr = df[['SALES','QUANTITYORDERED','PRICEEACH']].corr()
corr
```

Out[27]:

	SALES	QUANTITYORDERED	PRICEEACH
SALES	1.000000	0.536213	0.664905
QUANTITYORDERED	0.536213	1.000000	0.001639
PRICEEACH	0.664905	0.001639	1.000000